David Ulises Trejo Cancino

Senior Environmental Monitoring Technician | Eddy Covariance and Greenhouse Gas Flux Networks

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Summary

Geophysicist and senior field technician with more than six years of experience deploying, operating and maintaining eddy covariance and meteorological monitoring networks in Canada and Chile. Currently leads the technical operations of a multi-institutional greenhouse gas (GHG) flux network in southern Quebec wetlands. Covers the full measurement chain: site installation, datalogger programming (CRBasic), off-grid power systems, preventive and corrective maintenance, QA/QC and data delivery to AmeriFlux. Supervises junior technicians and builds software tools for field operations.

Experience

- **Université du Québec à Montréal (UQAM)** — Montreal, QC, Canada
  Research Support Professional, Senior Technician — *Apr 2024 – Present*
  - Lead the technical operations of the Carbonique research program, a GHG monitoring network studying carbon dynamics in southern Quebec wetlands, with UQAM, McGill, Université de Montréal, Université Laval and UQTR; deploy and manage its eddy covariance flux towers and meteorological stations from installation to long-term operation.
  - Develop and maintain a single version-controlled CRBasic program for Campbell Scientific CR1000X dataloggers running at six stations, with site-specific configuration, sensor power switching and automatic low-voltage protection.
  - Perform preventive and corrective maintenance, calibration, and root-cause diagnosis on gas analyzers, sonic anemometers and biometeorological sensors.
  - Standardize field protocols and data acquisition workflows across the network, and maintain per-site QA/QC configurations in the CANFLUX Biomet.net data-cleaning pipeline.
  - Supervise and mentor junior technicians and interns, lead intern recruitment, and enforce safety protocols.
  - Manage equipment inventory and procurement, coordinate field logistics across sites and research teams, and design 3D-printed field components (e.g., soil collars for GHG chamber measurements).

- **Carbon Real** — Santiago, Chile (remote)
  Carbon Fluxes Scientist — *Apr 2024 – Present*
  - Advise on eddy covariance station installation, and process and analyze flux data to quantify ecosystem carbon exchange.
  - Troubleshoot technical issues with colleagues and contribute to best practices for deploying and managing flux stations in diverse environments.

- **Instituto de Ecología y Biodiversidad (IEB Chile)** — Chile
  Technical Specialist, Long Term Socio-Ecological Research Network (LTSER) — *Jan 2023 – Mar 2024*
  - Managed meteorological and eddy covariance stations of the LTSER network and processed their data.

  Consultant, Data Infrastructure — *Jun 2022 – Dec 2022*
  - Defined requirements for a data warehouse server receiving high-volume station data, and built a visualization web app to detect instrument anomalies.

- **UMAG, Cape Horn International Center (CHIC)** — Magallanes Region, Chile
  Technical Specialist — *Apr 2022 – Mar 2024*
  - Managed and processed eddy covariance and soil respiration data (ANID/BASAL FB210018) and acted as meteorological consultant.

- Installed eddy covariance flux towers and automated soil respiration systems at Parque Omora, Puerto Williams, within FONDEQUIP project EQM200198, a GHG monitoring network for the world's southernmost forests and peatlands (May 2022).

- **Universidad de Chile, Ecosystem Ecology Laboratory (LECS)** Santiago, Chile
  Data Manager, Field and Laboratory Technician *Jul 2021 – Mar 2024*
  - Acquired, processed and managed soil respiration, biometeorological and eddy covariance data from high-frequency towers (FONDECYT project 1211652).
  - Grew responsibility to six eddy covariance stations; first in Chile to submit station data to AmeriFlux and the PhenoCam Network.
  - Built statistical and machine learning models of ecosystem fluxes, contributing to peer-reviewed publications.

- **INIA Intihuasi (Agricultural Research Institute)** Chile
  Consultant *Apr 2023*
  - Programmed and deployed a Surface Renewal station to account for the energy flux of an agricultural ecosystem and study its evapotranspiration.

- **Universidad de Chile and (CR)2** Chile
  Research Assistant and Research Intern *Jan 2018 – Jun 2021*
  - Investigated the mechanisms driving precipitation variability in Western Patagonia using statistics and clustering analysis (machine learning) of station and ERA5 reanalysis data (FONDECYT 11190952; FONDAP 15110009).
  - Installed two meteorological stations and two dust collectors at the Cerro Dominador solar complex and the ALMA observatory, Atacama Region, and supported campaign logistics (FONDECYT 1181139, 2020).
  - Performed statistical analysis of chemical traces on Rapa Nui (Easter Island) to interpret ozone soundings, linking volatile organic compounds to tropospheric ozone increases.

Technical Skills

- **Environmental Instrumentation**: Eddy covariance systems (LI-COR and Campbell Scientific equipment), meteorological stations, radiation sensors, precipitation gauges, soil respiration and GHG chamber systems
- **Data Acquisition and Control**: Scientific dataloggers, CRBasic, Modbus TCP, relay and power control, remote station access
- **Power Systems**: Off-grid solar PV, battery banks, methanol fuel cells (SFC EFOY), low-voltage protection
- **Flux Data Processing and QA/QC**: EddyPro, Biomet.net, time-series analysis, gap-filling, diagnostics, AmeriFlux and PhenoCam data submission
- **Programming and Software**: Python (NumPy, pandas, SciPy, Matplotlib, Plotly, Dash, scikit-learn, TensorFlow), MATLAB, Bash, Linux, Git, GitHub Actions, HPC (SLURM), AI Generative tools.
- **Field Operations**: Multi-site logistics, inventory and procurement, team supervision, work at heights, 3D design and printing of field parts.

Projects

- **Carbonique Field Log**: Web app for the monitoring network that tracks per-site tasks, field notes and visit history (GitHub Pages, Firebase). davidulicio.github.io/carbonique-field-log
- **EC Monitor**: Android app for read-only status checks of LI-COR gas analyzers over TCP, with releases built by GitHub Actions. github.com/davidulicio/EC-Monitor

Education

- **Universidad de Chile** Santiago, Chile
  Bachelor of Science in Geophysics

Publications and Datasets

Peer-Reviewed Articles

- Fuentes-Castillo, T., Grau-Neira, A., Morales-Santana, E., Rus-Valledor, D., **Trejo-Cancino, D.**, Pascual, A., and Perez-Quezada, J. F. (2026). An integrative methodology to estimate high-resolution carbon stock and fluxes: a case study in the old-growth forests of the Chilean Patagonia. *Carbon Balance and Management*, 21, 11. doi.org/10.1186/s13021-025-00381-6
- Reed, D. E., Chu, H., Peter, B. G., et al., including **Trejo-Cancino, D.** (2025). Network of networks: Time series clustering of AmeriFlux sites. *Agricultural and Forest Meteorology*, 372, 110686. doi.org/10.1016/j.agrformet.2025.110686
- Perez-Quezada, J. F., **Trejo, D.**, Lopatin, J., Aguilera, D., Osborne, B., Galleguillos, M., Zattera, L., Celis-Diez, J. L., and Armesto, J. J. (2024). Comparison of carbon and water fluxes and the drivers of ecosystem water use efficiency in a temperate rainforest and a peatland in southern South America. *Biogeosciences*, 21, 1371–1389. doi.org/10.5194/bg-21-1371-2024

Datasets

- Zimmerman, O., Young, A. M., Milliman, T., et al., including **Trejo-Cancino, D.** (2025). PhenoCam Dataset v3.0: Vegetation Phenology from Digital Camera Imagery, 2000–2023. ORNL DAAC. doi.org/10.3334/ORNLDAAC/2389
- Ballou, K., Vladich, Z., Young, A. M., et al., including **Trejo-Cancino, D.** (2025). PhenoCam Dataset v3.0: Digital Camera Imagery from the PhenoCam Network, 2000–2023. ORNL DAAC. doi.org/10.3334/ORNLDAAC/2364

Conference Contributions

- Garneau, M., del Giorgio, P., Davidson, S., et al., including **Trejo Cancino, D.** (2026). The CARBONIQUE project: Carbon cycling in Quebec's wetlands. Oral presentation, *EGU General Assembly 2026*, EGU26-17755. doi.org/10.5194/egusphere-egu26-17755
- Wang, T., Senay, G. B., Verfaillie, J. G., et al., including **Trejo-Cancino, D.** (2025). The category is water, water everywhere, and with it arrives the Variance-bOwen Ratio Terrestrial Evapotranspiration fluX (VORTEX) network. Oral presentation, *AGU Fall Meeting 2025*, abstract H41D-06.
- Fuentes-Castillo, T., Pérez-Quezada, J., Escalona, F., et al., including **Trejo, D.** (2025). Carbon Real: a land conservation model in which nature benefits from carbon markets and highly reliable greenhouse gas measurements. Oral presentation, *EGU General Assembly 2025*, EGU25-20617. doi.org/10.5194/egusphere-egu25-20617
- Perez-Quezada, J., **Trejo, D.**, Lopatin, J., et al. (2025). Ecosystem water use efficiency in a forest and a peatland in northern Patagonia. Poster, *EGU General Assembly 2025*, EGU25-2630. doi.org/10.5194/egusphere-egu25-2630
- **Trejo, D.** (presenter), and Boisier, J. P. (2023). Large-scale control of precipitation in Northwestern Patagonia. Oral presentation, *SICyR International Symposium: Climate Change and Resilience in Changing Times.*
- **Trejo Cancino, D.**, Pérez-Quezada, J., Lara, A., Thalasso, F., and Squeo, F. (2023). Micrometeorological drivers of carbon and water fluxes in natural ecosystems in Chile. Poster, *SICyR International Symposium: Climate Change and Resilience in Changing Times.*
- Boisier, J. P., and **Trejo, D.** (2022). Circulation and precipitation changes in the extratropical Southeast Pacific region: Reducing the uncertainties in climate model simulations. Poster, *AGU Fall Meeting 2022*, abstract A52L-1116.

Certifications, Training and Languages

- **Certifications**: Work at Heights; Python 101 for Data Science (IBM); Photovoltaic Energy Course; EF SET English C2 (83/100)
- **Training**: Deep Learning for Time Series and Image Classification Workshop, NLHPC Chile, 2022
- **Languages**: Spanish (native), Portuguese (bilingual), English (professional working proficiency)